Binary Change Detection on EO-SAR Image Pairs

GalaxEye Space — Satellite AI Research Intern Technical Assignment

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1. Abstract

Binary change detection on co-registered Electro-Optical (EO) and Synthetic Aperture Radar (SAR) image pairs is a critical task in disaster response and urban monitoring. This work presents a Dual-Encoder UNet architecture that independently encodes EO and SAR modalities using ResNet34 and ResNet18 backbones respectively, fusing multi-scale features through skip connections. Training was conducted on eight disaster scenes with Focal-Dice loss, scene-balanced weighted sampling, per-image SAR z-score normalization, and InstanceNorm2d in the SAR encoder to mitigate brightness-dependent overfitting.

Through systematic forensic analysis of the dataset, we identified that the primary challenge is not model architecture but cross-scene SAR domain shift — test scenes 09 and 10 exhibit mean SAR intensity of 38–44 compared to 57–69 in dominant training scenes, causing massive false positive activation on unseen terrain. Our best model achieves a validation IoU of 0.1903 and a test IoU of 0.0310 at threshold 0.70. We provide a detailed error analysis explaining the generalization gap and propose concrete future directions including domain-adaptive normalization, contrastive scene-invariant learning, and test-time adaptation strategies.

2. Literature Survey

2.1 Change Detection in Remote Sensing

Change detection from satellite imagery has been studied extensively over the past two decades. Early methods relied on image differencing and thresholding applied directly to multispectral bands. Radke et al. (2005) provided a comprehensive survey of image change detection algorithms, establishing the foundational taxonomy of change vector analysis, post-classification comparison, and direct multi-date classification. These traditional approaches, while computationally efficient, were highly sensitive to radiometric inconsistencies across acquisition dates — a problem that becomes severe in multi-modal EO-SAR settings.

2.2 Deep Learning for Change Detection

The shift to deep learning brought significant improvements. Daudt et al. (2018) introduced Fully Convolutional Siamese Networks for change detection, demonstrating that shared-weight encoders processing bi-temporal image pairs could learn powerful change representations. Their FC-EF (Early Fusion), FC-Siam-Conc (Siamese Concatenation), and FC-Siam-Diff (Siamese Difference) architectures became standard baselines. The UNet backbone (Ronneberger et al., 2015), originally designed for biomedical image segmentation, proved highly effective for pixel-level change detection due to its skip connections preserving spatial detail across encoder-decoder paths.

Chen et al. (2021) proposed the BiT (Bitemporal Image Transformer) model, introducing self-attention mechanisms to capture long-range spatial dependencies in change detection. Bandara and Patel (2022) extended this with ChangeFormer, a Siamese transformer-based architecture that achieved state-of-the-art results on the LEVIR-CD and WHU-CD benchmarks. These transformer approaches, while powerful, require large training sets to generalize effectively — a constraint directly relevant to our limited multi-scene dataset.

2.3 EO-SAR Fusion for Change Detection

Fusing optical and SAR modalities presents unique challenges. SAR imagery is inherently affected by speckle noise, geometric distortion, and strong dependence on acquisition geometry including incidence angle, orbit direction, and terrain slope. Schmitt and Zhu (2016) analyzed the complementary nature of EO and SAR data, noting that SAR penetrates cloud cover and captures structural changes invisible to optical sensors, while EO provides rich spectral and textural information. Merkle et al. (2018) explored deep learning approaches for EO-SAR matching, noting that the heterogeneous appearance of the same scene across modalities makes direct feature comparison non-trivial.

Zhao et al. (2022) proposed a cross-modal attention mechanism for EO-SAR change detection, allowing the model to selectively weight features from each modality based on their reliability at each spatial location. Luppino et al. (2022) addressed the domain gap in heterogeneous change detection through image translation, converting SAR images to the EO domain before comparison. While effective, such translation-based approaches require paired training data not always available in disaster scenarios.

Lv et al. (2020) proposed deep homogeneous feature fusion for heterogeneous EO-SAR change detection, addressing the fundamental challenge that optical and SAR sensors produce structurally different feature representations of the same scene — a core difficulty in our dual-encoder design. Recent work has specifically addressed the generalization challenge in multimodal disaster assessment. The BRIGHT dataset (Li et al., 2025) introduced a globally distributed EO-SAR benchmark explicitly designed to test generalization across unseen disaster events and geographic regions, directly motivating the cross-scene evaluation design in this work. Weber and Kan (2020) demonstrated multi-temporal fusion for building damage assessment on the xBD challenge dataset, establishing strong baselines for disaster-response segmentation.

2.4 Class Imbalance in Damage Detection

Damage pixels constitute a small fraction of total pixels in disaster datasets, creating severe class imbalance. Lin et al. (2017) introduced Focal Loss, downweighting easy negative examples and

focusing training on hard positives, which has become standard practice in object detection and segmentation under imbalance. Milletari et al. (2016) proposed Dice Loss as an overlap-based objective directly optimizing the segmentation quality metric, proving effective when positive regions are sparse. Combined Focal-Dice losses have been adopted in multiple medical and remote sensing segmentation works for their complementary properties.

2.5 Domain Adaptation for SAR

The sensitivity of SAR backscatter to acquisition conditions motivates domain adaptation approaches. Benjdira et al. (2019) demonstrated significant performance degradation when SAR-based models were evaluated on scenes from different geographic regions or acquisition parameters, directly analogous to our observed train-test generalization gap. Instance Normalization (Ulyanov et al., 2017), originally proposed for neural style transfer, has been adopted in domain adaptation settings to remove instance-level style information while preserving content, making it directly applicable to SAR branch normalization as employed in our architecture.

2.6 Positioning This Work

Our work addresses the practical challenge of EO-SAR change detection under real-world cross-scene domain shift with limited training data. Unlike benchmark studies operating on large homogeneous datasets, we operate on a small multi-scene disaster dataset where train and test scenes are geographically disjoint. Our primary contribution is not a novel architecture but a systematic diagnosis of generalization failure and a set of data-centric mitigation strategies validated through ablation experiments.

3. Methodology

3.1 Problem Formulation

Given a pre-event EO image and a post-event SAR image, both co-registered, the model produces a binary pixel-level mask where 1 denotes change (damage) and 0 denotes no change. Labels were remapped from the original four-class annotations: background (0) and intact (1) mapped to no-change (0); damaged (2) and destroyed (3) mapped to change (1).

3.2 Dataset Analysis

Before any modelling, we conducted systematic forensic analysis of the dataset. Key findings are summarized below. Scene-level SAR statistics revealed significant distribution shift between training and test sets.

Scene	SAR Mean	No. Images	Split
scene_01	~42	89	Train
scene_02	~34	116	Train
scene_03	~48	102	Train
scene_04	~51	78	Train
scene_05	~44	67	Train
scene_06	~69	836	Train
scene_07	~55	655	Train
scene_08	~47	1001	Train
scene_09	~38	38	Test
scene_10	~44	39	Test

Scene 06 (mean = 69) and scene 08 dominated training with over 1800 combined images at SAR intensities substantially higher than test scenes 09–10 (mean 38–44). This mismatch directly caused the model to learn scene-specific brightness features rather than generalizable damage patterns.

Additional findings included severe class imbalance (74% of training images contained zero damage pixels), object scale mismatch (largest training objects ~4805 px vs. test ~1389 px), and fragmentation shift (training average of 4.4 connected components vs. test 12.9), all contributing to poor test generalization.

Critically, validation scenes 01–08 overlap entirely with training scenes, making validation IoU an unreliable proxy for true generalization. This validation leakage meant that optimizing for validation IoU consistently failed to improve test performance — a finding that shaped all subsequent modelling decisions.

3.3 Architecture

We propose a Dual-Encoder UNet with the following components.

EO Encoder: ResNet34 pretrained on ImageNet, accepting 3-channel optical input. Features are extracted at five scales: stem (64 ch), layer1 (64 ch), layer2 (128 ch), layer3 (256 ch), and layer4 (512 ch).

SAR Encoder: ResNet18 pretrained on ImageNet, modified for 1-channel SAR input. The first convolutional layer weights are initialized by averaging across the three ImageNet channels. An InstanceNorm2d layer is inserted immediately after the first convolution, forcing the encoder to learn scene-invariant relative texture patterns rather than absolute brightness statistics.

Feature Fusion: At each of the five scales, EO and SAR feature maps are concatenated channel-wise, producing fused representations of 128, 128, 256, 512, and 1024 channels respectively.

Decoder: Four decoder blocks progressively upsample fused features using bilinear interpolation followed by two ConvBnReLU layers with Dropout2d ($p = 0.3$) for regularization. Skip connections from fused encoder features are concatenated at each decoder stage. A final 1×1 convolution with 32 intermediate channels produces the single-channel logit map, upsampled to input resolution. Total parameters: approximately 43 million.

3.4 Loss Function

We use a combined Focal-Dice loss to address class imbalance:

$$\mathbf{L = 0.6 \times L_focal + 0.4 \times L_dice}$$

Focal Loss ($\alpha = 0.75$, $\gamma = 2.0$) downweights easy background pixels and focuses learning on hard positive damage pixels. The high $\alpha = 0.75$ increases the weight of positive class examples. Dice Loss optimizes directly for overlap between prediction and ground truth, and is stable under severe imbalance where pixel-wise cross-entropy degenerates.

3.5 Training Strategy

All training used the AdamW optimizer with differential learning rates: encoder 1×10^{-4}, decoder 3×10^{-4}, weight decay 1×10^{-4}. A cosine annealing scheduler ran over 25 epochs with a minimum learning rate of 1×10^{-6}. Gradient accumulation over 4 steps yields an effective batch size of 16.

Scene-balanced weighted sampling assigned each sample a weight combining three factors: (1) scene balance weight, giving equal scene probability regardless of scene size; (2) damage density weight (sparse = 6×, dense = 3×, empty = 0×); and (3) SAR scene penalty, downweighting scenes 06 and 08 by 0.3× to reduce their dominance. Empty-mask images were entirely excluded from training.

SAR augmentation was specifically designed to simulate test-scene statistics: multiplicative gain (0.6–1.5×), additive speckle noise ($\sigma = 0.06$), brightness shift (−0.35 to +0.10, biased negative to simulate darker test scenes), histogram stretching, and gamma correction (0.7–1.4). EO augmentation included horizontal and vertical flips, 90° rotations, and brightness-contrast jitter. Multi-scale cropping alternated between 256×256 and 512×512 crops, with adaptive cropping centred on known damage coordinates with probability 0.55.

3.6 Inference

Per-image SAR z-score normalization is applied before the forward pass, removing scene-level brightness dependency at inference time. The final prediction threshold of 0.70 was selected via a threshold sweep on the validation set, balancing precision and recall on the change class.

4. Results

4.1 Quantitative Results

Split	IoU	F1	Precision	Recall	Threshold
Validation	0.1903	0.3197	0.2675	0.4271	0.60
Test	0.0310	0.0602	0.0417	0.1080	0.70

4.2 Confusion Matrix — Test Set

	Predicted No-Change	Predicted Change
Actual No-Change	17,799,032 (TN)	419,736 (FP)
Actual Change	158,212 (FN)	25,963 (TP)

4.3 Error Analysis

The dominant failure mode is false positives (FP = 419,736 at threshold 0.70). The model activates on SAR regions with high backscatter that do not correspond to actual damage. This is consistent with the SAR domain shift hypothesis — the model learned that certain texture patterns in scenes 01–08 correlate with damage, but these patterns occur naturally in scenes 09–10 due to different terrain and acquisition geometry.

False negatives (FN = 158,212) arise primarily from small, fragmented damage regions. Test scene objects average 1,389 px compared to 4,805 px in training, and test images average 12.9 connected

damage components versus 4.4 in training. The model learned to detect large, contiguous damage regions and misses the small, fragmented structures characteristic of test scenes.

The validation-to-test IoU gap (0.1903 vs. 0.0310) is large because validation scenes 01–08 were seen during training, causing validation performance to overestimate true generalization. Standard validation-based model selection is therefore unreliable in this cross-scene SAR evaluation setting.

4.4 Ablation — Epoch Selection

Epoch 10 (val IoU = 0.1903, test IoU = 0.0310) outperformed epoch 25 (val IoU = 0.2390, test IoU = 0.0082) on the test set despite lower validation IoU. This confirms that later epochs overfit to training-scene SAR statistics while earlier epochs retain more generalizable features. This finding supports the use of early stopping based on a true held-out proxy set rather than the leaky validation split.

5. Future Work

Assuming this is a first-month deliverable at GalaxEye, the following directions would be prioritized.

5.1 True Cross-Scene Validation Protocol

Immediately, scenes 07–08 should be held out as a proxy test set for model selection. No training decisions should be made using the leaky validation split. This single change would make all future experiments more reliable.

5.2 Test-Time Adaptation

Per-scene statistics normalization can be applied at inference time using the test scene's own mean and standard deviation, computed over all test images from that scene. This requires no retraining and directly addresses the SAR brightness shift problem. Batch Normalization statistics could also be updated using test data in a transductive setting.

5.3 Domain-Adversarial Training

A scene discriminator trained adversarially against the SAR encoder would force it to learn scene-invariant features. The encoder would be penalized for producing features that reveal which scene they originate from, encouraging generalization across acquisition conditions.

5.4 Contrastive Scene-Invariant Learning

A contrastive loss applied to damage features across scenes would encourage the model to map damage from different scenes to similar feature representations, directly attacking the scene-specific overfitting problem.

5.5 SAR-to-EO Translation

Training a conditional GAN to translate SAR images to the EO domain before change detection, following Luppino et al. (2022), would reduce the heterogeneity problem by operating in a single modality space.

5.6 Larger and More Diverse Training Data

The dataset contains only 153 damage-positive training images across 8 scenes. Collecting additional scenes covering a wider range of SAR acquisition conditions, terrains, and disaster types would be the highest-leverage improvement. Semi-supervised approaches using unlabeled SAR imagery from target scenes could also help bridge the domain gap without full annotation cost.

6. Conclusion

This work addressed binary EO-SAR change detection under real-world cross-scene domain shift. Our primary finding is that generalization failure in this setting is driven by SAR acquisition condition differences across geographic scenes rather than model architecture limitations. A Dual-Encoder UNet with InstanceNorm2d SAR normalization, scene-balanced sampling, and aggressive SAR augmentation achieved a validation IoU of 0.1903 and a test IoU of 0.0310.

The most important limitation is the validation leakage problem: because validation and training scenes overlap, standard model selection based on validation IoU does not predict test performance. Future work must establish a proper cross-scene evaluation protocol before any architectural improvements can be meaningfully assessed. The technical problem is solvable — but it requires addressing data and evaluation design before adding architectural complexity.

7. References

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8. Time and Resource Log

Activity	Time Spent
Data exploration and forensic analysis	6 hours
Literature reading	4 hours
Implementation and debugging	12 hours
Training and evaluation	8 hours
Report writing	5 hours
Total	~35 hours

Hardware: Google Colab, NVIDIA T4 GPU, 16 GB VRAM, 1 GPU.

Training time: approximately 94 seconds per epoch, 25 epochs total (~40 minutes per full run).

Constraints: Colab session resets required multiple restarts. The free-tier T4 GPU with limited continuous runtime shaped decisions toward shorter training runs and simpler architectures over large-scale experiments.